

Clustering Methods

Hierarchical, Sequential, Mixture Models, *K*-Means, Fuzzy *K*-Means,
and Self-Organizing Maps

Prof. Dr. Rogério Galante Negri

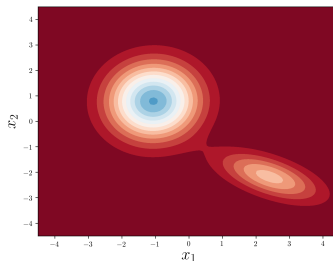
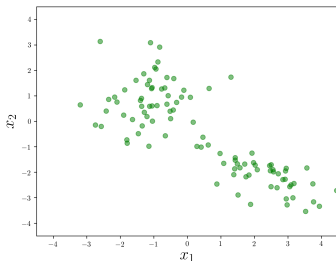
Agenda

- 1 Data Clustering
- 2 Sequential Clustering
- 3 Clustering via Mixture Models
- 4 Clustering via Sum of Squares
- 5 Computational Experiment

Data Clustering

Definition and Concept

- **Clusters** are regions in the feature space that contain high densities of patterns and are separated by regions of low density.
- **Clustering methods** encompass data exploration techniques that reveal relationships emerging from the data.
- The properties of a cluster can be used to summarize the behavior of the corresponding dataset.



Data Clustering

Motivations

- It is reasonable to assume that patterns within the same cluster exhibit similarity among themselves and dissimilarity with respect to patterns in other clusters.
- Key elements in the clustering process include:
 - Dissimilarity measures
 - Aggregation criteria — rules that determine how patterns are grouped together to form clusters
- The literature on this topic is extensive, and its applications are diverse (signal processing, psychology, archaeology, linguistics, ...)
- Before starting, it is important to understand the problem and the data.
- Different clustering methods can lead to different interpretations (choose wisely!)

Unsupervised paradigm, remember?

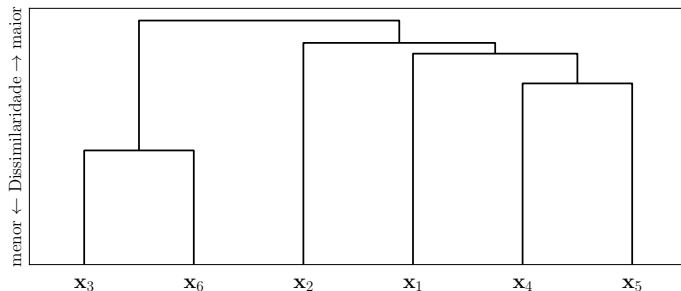
- Clustering methods – **unsupervised classification methods**
- They can be expressed as $g: \mathcal{I} \rightarrow \mathcal{G}$, where $\mathcal{I} \subseteq \mathcal{X}$ is a set of m examples $\mathbf{x}$ observed in the attribute space $\mathcal{X}$, and $\mathcal{G} = \{\mathcal{G}_1, \dots, \mathcal{G}_c\}$ is a partition of $\mathcal{I}$ into c subsets
- No knowledge is available regarding the labels of the examples in $\mathcal{I}$
- It is ensured that: $\mathcal{G}_j \neq \emptyset$; $\bigcup_{j=1}^c \mathcal{G}_j = \mathcal{I}$; and $\bigcap_{j=1}^c \mathcal{G}_j = \emptyset$ – this defines a **hard clustering**, as each example belongs to a single cluster
- There is also **fuzzy clustering**, based on the concept of fuzzy sets – the patterns are assigned to all clusters according to their **degree of membership**

Categories

- The following will be addressed:
 - Hierarchical
 - Single-link
 - Complete-link
 - Average-link
 - Ward's method
 - Sequential
 - Mixture Models
 - Cost Function Optimization-Based
 - *K*-Means (Hard)
 - Fuzzy *K*-Means
 - Others — Competitive
 - Kohonen Self-Organizing Maps

Hierarchical Clustering

- Hierarchical methods are commonly used to synthesize the organizational structure by which data are related to one another.
- Based on the dissimilarity values between elements, the existence of subdivisions becomes evident with respect to a given threshold.
- These subdivisions naturally determine the configuration of the clusters.



Hierarchical Clustering

- **Agglomerative hierarchical** approaches proceed by consecutively merging clusters in the dataset until a single cluster is reached.
- **Divisive hierarchical** algorithms follow the reverse path (not covered here).
- The **dissimilarity** measure is the basis for comparison.
- Euclidean, cosine, or Manhattan distances are commonly used.
- We denote by $d(\mathbf{x}_i, \mathbf{x}_k)$ the dissimilarity between patterns $\mathbf{x}_i$ and $\mathbf{x}_k$.
- $d(\mathbf{x}_i, \mathbf{x}_k) \rightarrow \infty$ indicates greater dissimilarity.
- $d(\mathbf{x}_i, \mathbf{x}_k) \rightarrow 0$ indicates greater similarity.
- $D(\mathcal{G}_j, \mathcal{G}_\ell)$ denotes the dissimilarity between clusters $\mathcal{G}_j$ and $\mathcal{G}_\ell$.

Dissimilarities and Derived Methods

Cluster formation is established through dissimilarity between clusters.

Single-link – SL

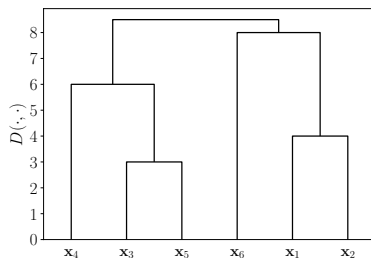
$$D(\mathcal{G}_j, \mathcal{G}_\ell) = \min_{\mathbf{x}_a \in \mathcal{G}_j; \mathbf{x}_b \in \mathcal{G}_\ell} d(\mathbf{x}_a, \mathbf{x}_b) \quad (1)$$

Complete-link – CL

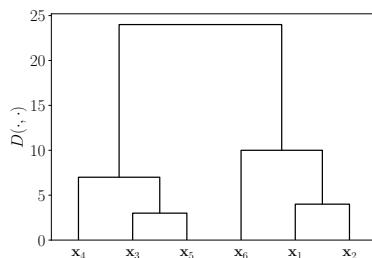
$$D(\mathcal{G}_j, \mathcal{G}_\ell) = \max_{\mathbf{x}_a \in \mathcal{G}_j; \mathbf{x}_b \in \mathcal{G}_\ell} d(\mathbf{x}_a, \mathbf{x}_b) \quad (2)$$

We assume, as a starting point, that each cluster is composed of a single pattern (i.e., $\mathcal{G}_i = \{\mathbf{x}_i\}$, for $i = 1, \dots, m$) — for a cluster to expand, it is sufficient that any of its elements exhibits a dissimilarity below a given threshold.

Dissimilarities and Derived Methods (cont.1)



(a) SL



(b) CL

Figure: Dendrograms obtained using the SL and CL methods.

While SL yields clusters that are separated from each other by a minimum distance τ , the clusters defined by CL have, at most, a dissimilarity equal to τ .

Dissimilarities and Derived Methods (cont.2)

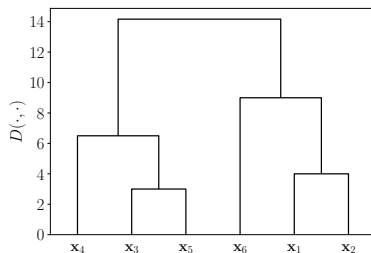
Average-link – AL

$$D(\mathcal{G}_j, \mathcal{G}_\ell) = \frac{1}{\#\mathcal{G}_j \#\mathcal{G}_\ell} \sum_{\mathbf{x}_a \in \mathcal{G}_j, \mathbf{x}_b \in \mathcal{G}_\ell} d(\mathbf{x}_a, \mathbf{x}_b) \quad (3)$$

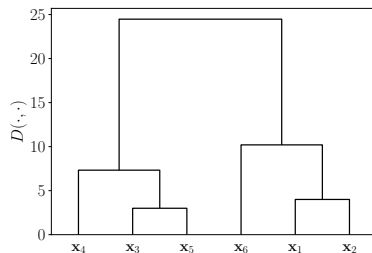
Ward

$$D(\mathcal{G}_j \cup \mathcal{G}_k, \mathcal{G}_\ell) = \frac{\#\mathcal{G}_j + \#\mathcal{G}_\ell}{\#\mathcal{G}_j + \#\mathcal{G}_k + \#\mathcal{G}_\ell} D(\mathcal{G}_j, \mathcal{G}_\ell) + \frac{\#\mathcal{G}_k + \#\mathcal{G}_\ell}{\#\mathcal{G}_j + \#\mathcal{G}_k + \#\mathcal{G}_\ell} D(\mathcal{G}_k, \mathcal{G}_\ell) - \frac{\#\mathcal{G}_\ell}{\#\mathcal{G}_j + \#\mathcal{G}_k + \#\mathcal{G}_\ell} D(\mathcal{G}_j, \mathcal{G}_k) \quad (4)$$

Dissimilarities and Derived Methods (cont.3)



(a) AL



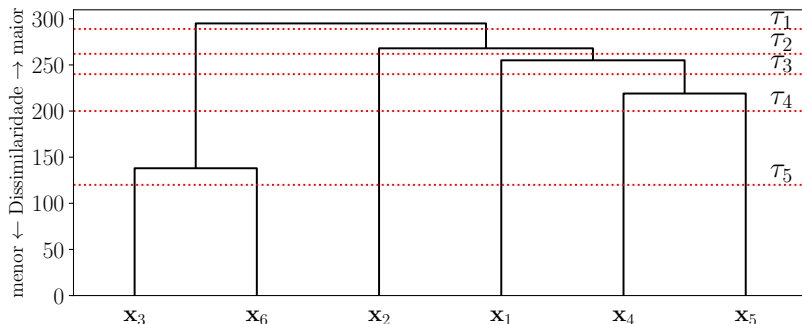
(b) Ward

Figura: Dendrograms obtained using the AL and Ward methods.

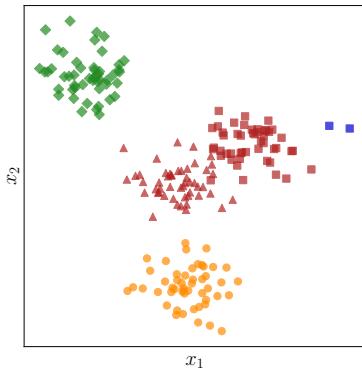
In this example, the difference between such methods lies only in the dissimilarity values required to define the clusters.

Hierarchy and Cluster Definition

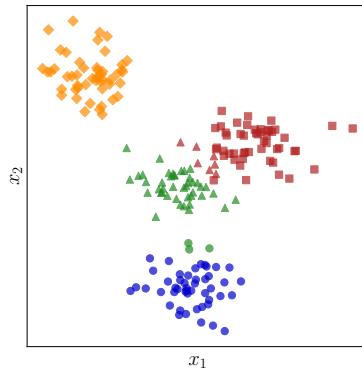
- The discussions presented so far reveal the hierarchical structure within the data.
- The partitioning of the data into different clusters can be achieved by applying a threshold starting from the highest observed dissimilarity value and decreasing it to the next lowest levels.



Comparison Between Methods (SL left and CL right)

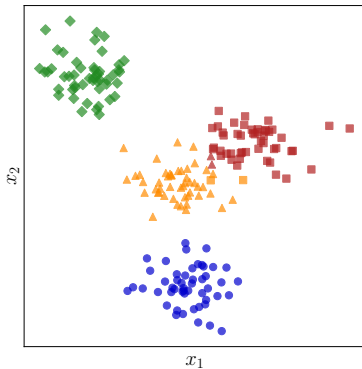


(a) SL

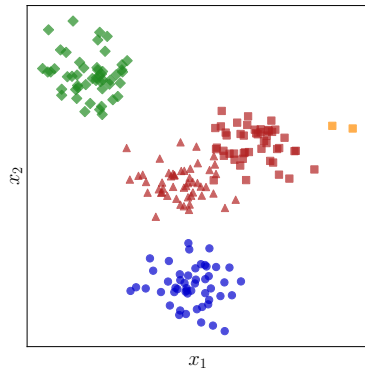


(b) CL

Comparison Between Methods (Ward and AL)



(c) Ward



(d) AL

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Sequential Clustering

- A direct and simple way to define a partition over the dataset.
- The patterns are presented a few times, following a certain sequence
- There are three main elements:
 - 1 Dissimilarity measure $d_j(\mathbf{x}_i)$ between $\mathbf{x}_i$ and cluster $\mathcal{G}_j$;
 - 2 Threshold τ used to assess dissimilarity values;
 - 3 Maximum number of clusters allowed, defined as c .

BSAS – *Basic Sequential Algorithm Scheme*

- The number of clusters is not predefined, but determined during execution.
- The process starts by defining a cluster composed of the first example observed in the sequence $\mathcal{I} = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$.
- Subsequent examples are presented to this cluster, which may incorporate them if the dissimilarity is below τ .
- Otherwise, a new cluster is initiated with the current example.

Sequential Clustering

BSAS – pseudocode

- 1 Define τ, c
- 2 Initialize the counter $\ell \leftarrow 1$
- 3 Set $\mathcal{G}_\ell = \{\mathbf{x}_1\}$
- 4 For $i = 2, \dots, m$, do:
 - 1 $k = \underset{j=1, \dots, \ell}{\operatorname{argmin}} d_j(\mathbf{x}_i)$
 - 2 If $d_k(\mathbf{x}_i) > \tau$ and $\ell < c$, then:
 - 1 $\ell \leftarrow \ell + 1$
 - 2 $\mathcal{G}_\ell = \{\mathbf{x}_i\}$
 - 3 $\bar{\mu}_\ell = \mathbf{x}_i$
 - 3 Else:
 - 1 $\mathcal{G}_k \leftarrow \mathcal{G}_k \cup \{\mathbf{x}_i\}$
 - 2 $\bar{\mu}_k = \frac{(\#\mathcal{G}_k - 1)\bar{\mu}_k + \mathbf{x}_i}{\#\mathcal{G}_k}$

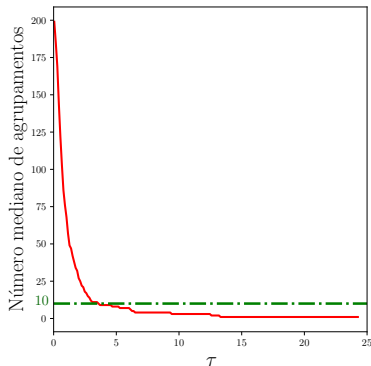
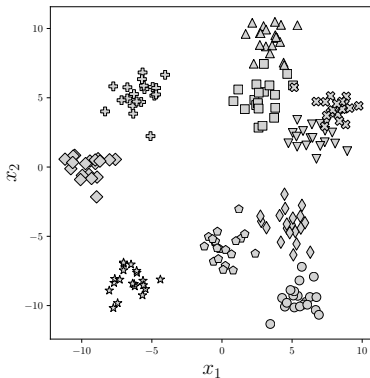
BSAS and the Number of Clusters

- Interesting procedure! – Evaluate different values of τ in the interval $[\tau_{min}, \tau_{max}]$ and observe the number of clusters formed (for different data presentation orders).
- For each value of τ considered, the median number of clusters identified is used to summarize the influence of this parameter.

```
1 # Exhaustive search for min/max distances
2 minTau, maxTau = find_minmax_tau(x)
3 # Number of reps. and evals. in the interval [minTau, maxTau]
4 repetitions = 10; steps = 200
5 vecTau = np.linspace(minTau, maxTau, steps)
6 vecClusters = []
7 for tau in vecTau:
8     counts = []
9     for _ in range(repetitions):
10         rand = np.argsort(np.random.randint(0, 1, x.shape[0]))
11         randX = np.copy(x[rand, :])
12         res = BSAS(x, tau, randX.shape[0])
13         counts.append(np.unique(res).shape[0])
14     vecClusters.append(np.median(counts))
```

BSAS and the Number of Clusters (cont.1)

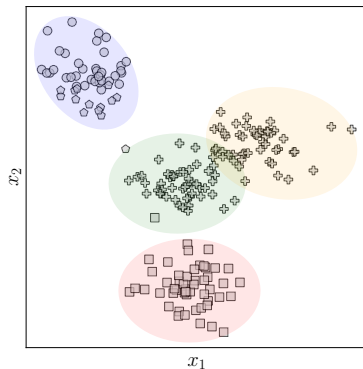
- A synthetic pattern set composed of ten clusters was used
- The number of detected clusters decreases rapidly for small values of τ , and then (for values close to 5 or higher) it tends to stabilize, exhibiting an asymptotic and stationary behavior



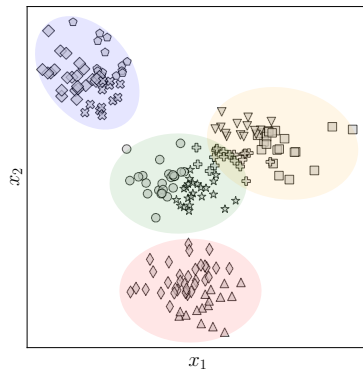
Modified BSAS

- MBSAS – Modified BSAS
- Aims to overcome an intrinsic weakness of the BSAS algorithm (which arises from the order in which patterns are presented)
- It is divided into two main steps:
 - 1 Identifies representatives for groups whose dissimilarity is greater than τ
 - 2 Based on the identified representatives, associates the remaining patterns with the cluster to which they are least dissimilar

BSAS and MBSAS Experiments

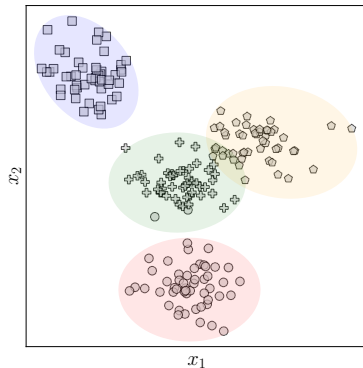


(e) BSAS; $c = 4$; $\tau = 2.0$

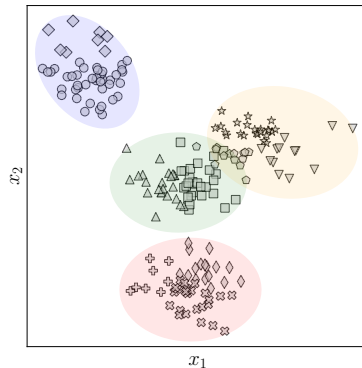


(f) BSAS; $c = 10$; $\tau = 1.0$

BSAS and MBSAS Experiments (cont.1)



(g) MBSAS; $c = 4$; $\tau = 2.0$



(h) MBSAS; $c = 10$; $\tau = 1.0$

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Clustering via Mixture Models

Gaussian Mixture Model (GMM)

A probability distribution $p(\mathbf{x})$ results from the linear combination of c other distributions, i.e.,

$$p(\mathbf{x}) = \sum_{j=1}^c \lambda_j p(\mathbf{x}|\theta_j), \quad (5)$$

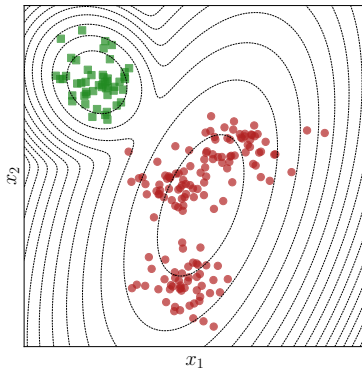
where θ_j is a tuple of parameters required by the j -th component, weighted by λ_j , for $j = 1, \dots, c$.

These parameters must satisfy: $\sum_{j=1}^c \lambda_j = 1$ and $\int_{\mathbf{x} \in \mathcal{X}} p(\mathbf{x}|\theta_j) d\mathbf{x} = 1$.

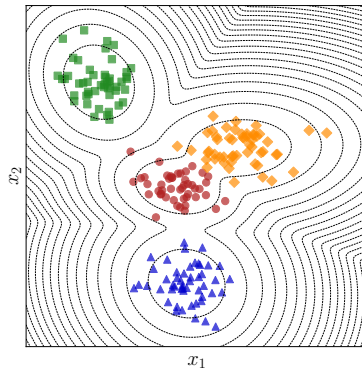
GMM revisited!

The terms in this linear combination represent the degree of membership of patterns to each of the c components. As a result, these terms can be used to define a clustering criterion as follows:

Clustering via GMM



(i) Two clusters



(j) Four clusters

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Clustering based on Sum of Squares

Clustering methods based on the “sum of squares” aim to find data partitions such that the internal variability within clusters is minimized while the separation between clusters is maximized.

■ Within-cluster variability

$$\mathbf{V}_I = \frac{1}{m} \sum_{j=1}^c \sum_{i=1}^m \delta(\mathbf{x}_i, \mathcal{G}_j) (\mathbf{x}_i - \mu_j)^T (\mathbf{x}_i - \mu_j) \quad (7)$$

■ Between-cluster separation

$$\mathbf{V}_E = \sum_{j=1}^c \frac{\#\mathcal{G}_j}{m} (\mu - \mu_j)^T (\mu - \mu_j) \quad (8)$$

Clustering criteria... based on the sum of squares

From the matrices $\mathbf{V}_I$ and $\mathbf{V}_E$, different clustering criteria are defined.
Two examples are:

- Minimization of $\text{Tr}(\mathbf{V}_I)$
- Maximization of $\text{Tr}(\mathbf{V}_I^{-1}\mathbf{V}_E)$

Given a clustering criterion, a **clustering algorithm** is employed to determine a partition over the data:

- *K*-Means (hard)
- Fuzzy *K*-Means

K-Means

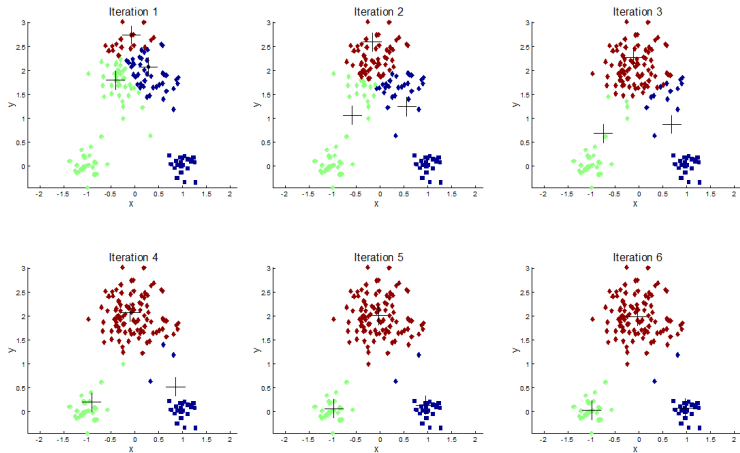
- The goal of this algorithm is to partition the set $\mathcal{I}$ into k clusters such that the within-cluster variability is minimized
- The adopted criterion is $\text{Tr}(\mathbf{V}_I)$, which reduces to the sum of squared deviations from each cluster centroid:

$$\text{Tr}(\mathbf{V}_I) = \frac{1}{m} \sum_{j=1}^c \sum_{\mathbf{x}_i \in \mathcal{G}_j} \|\mathbf{x}_i - \mu_j\|^2 \quad (9)$$

Iterative steps

- 1 Assign elements to the cluster whose centroid has the smallest Euclidean distance to the given sample
- 2 Recalculate the centroid of each cluster based on the elements assigned in the previous step

K-Means (process overview)



<http://www.practicaldb.com/data-visualization-consulting/cluster-analysis/attachment/k-means/>

Fuzzy K-Means

- Fuzzy K-Means (FCM) allows patterns to exhibit varying degrees of membership to the clusters
- The FCM algorithm seeks a configuration of parameters λ_{ij} that optimizes:

$$\min_{\substack{\lambda_{ij} \\ i=1,\dots,m \\ j=1,\dots,k}} \sum_{i=1}^m \sum_{j=1}^k \lambda_{ij} \|\mathbf{x}_i - \bar{\mu}_j\|^2 \quad (10)$$

where:

- λ_{ij} denotes the membership of $\mathbf{x}_i$ to cluster $\mathcal{G}_j$
- The constraints $\sum_{j=1}^k \lambda_{ij} = 1$ and $\lambda_{ij} \geq 0$ are enforced
- $\bar{\mu}_j = \frac{\sum_{i=1}^m \lambda_{ij}^\beta \mathbf{x}_i}{\sum_{i=1}^m \lambda_{ij}^\beta}$ represents the centroid of cluster $\mathcal{G}_j$
- $\beta \in]1, \infty[$ is an exponent that controls the fuzziness of the clusters, where $\beta \rightarrow 1$ leads to a **hard** clustering approach

Fuzzy K -Means (cont.)

Regarding the algorithm, the following steps are performed:

- 1 Define β and initialize the membership values λ_{ij} , for $i = 1, \dots, m$ and $j = 1, \dots, k$
- 2 Compute the centroids $\bar{\mu}_j$, for $j = 1, \dots, k$
- 3 Compute dissimilarities $d(\mathbf{x}_i, \bar{\mu}_j) = \|\mathbf{x}_i - \bar{\mu}_j\|$, for $i = 1, \dots, m$ and $j = 1, \dots, k$
- 4 Update the membership degrees via:

$$\lambda_{ij} = \sum_{j=1}^{\ell} \left(\frac{d(\mathbf{x}_i, \bar{\mu}_j) + \epsilon}{d(\mathbf{x}_i, \bar{\mu}_\ell) + \epsilon} \right)^{\frac{-2}{\beta-1}}, \quad (11)$$

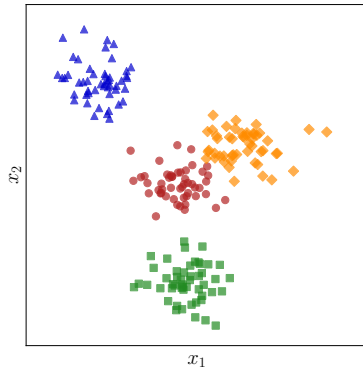
for $i = 1, \dots, m, j = 1, \dots, k$, where ϵ is an infinitesimal added to avoid numerical indeterminacy

- 5 If convergence has not been reached, return to step (ii)

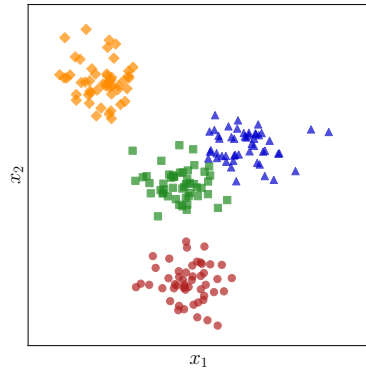
Convergence criterion – check the deviation between updated and previous centroids $\bar{\mu}_j$ and $\bar{\mu}_j^\dagger$, for $j = 1, \dots, k$:

$$\left[\sum_{j=1}^k \|\bar{\mu}_j - \bar{\mu}_j^\dagger\|^2 \right]^{\frac{1}{2}} \quad (12)$$

Comparison between hard and fuzzy K -Means – $k = 4$

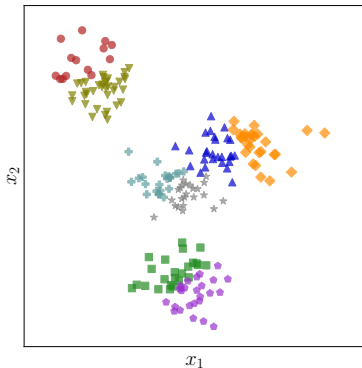


(k) KM

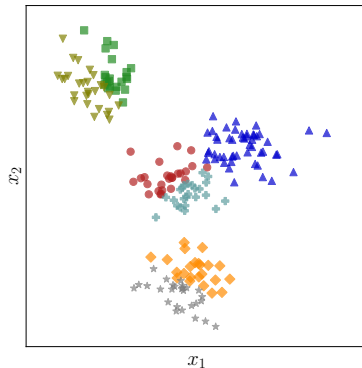


(l) FKM

Comparison between hard and fuzzy K -Means – $k = 8$



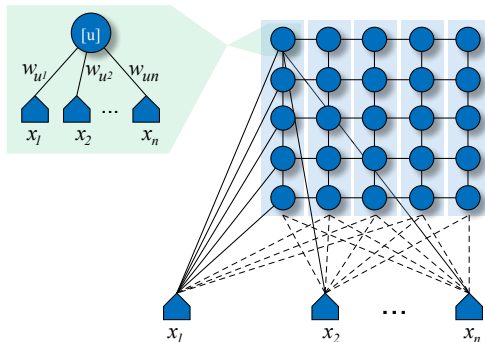
(m) KM



(n) FKM

Self-Organizing Maps (Kohonen)

- SOM – *Self-Organizing Maps*
- A neural network model inspired by the structural arrangement of the human brain
- Neurons are arranged in a topologically organized map, where the position/coordinates of the neurons encode specific characteristics of the input data



SOM

- The training of such a neural network involves three main processes:
competition, cooperation, and adaptation

Competition

Identifies the neuron with the smallest dissimilarity to the presented input:

$$[uv] = \arg \min_{\substack{u=1,\dots,L_1 \\ v=1,\dots,L_2}} \|\mathbf{x} - \mathbf{w}_{uv}\| \quad (13)$$

Cooperation

Adjustments are spread to the neighborhood of the winning neuron:

$$\mathcal{V}(\mathbf{a}, \mathbf{b}; \sigma) = e^{-\frac{\|\mathbf{a} - \mathbf{b}\|^2}{2\sigma^2}} \quad (14)$$

Adaptation

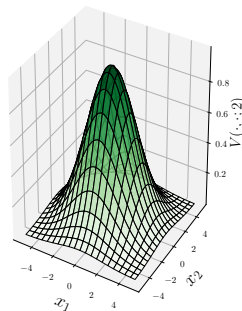
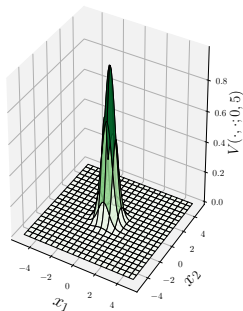
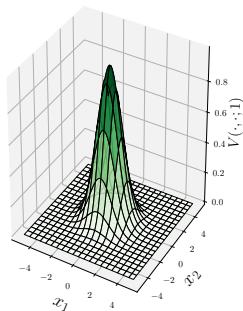
Updates the weight vectors as patterns are presented to the network:

$$\mathbf{w}_{ij} := \mathbf{w}_{ij} + \eta \mathcal{V}((i, j), (u, v), \sigma) (\mathbf{x} - \mathbf{w}_{ij}) \quad (15)$$

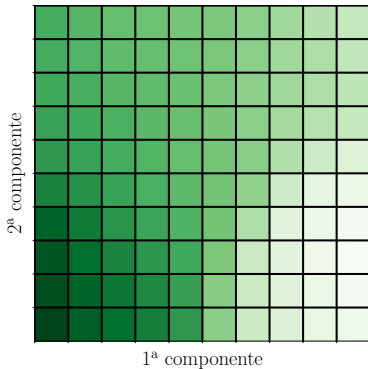


SOM – neighborhood function

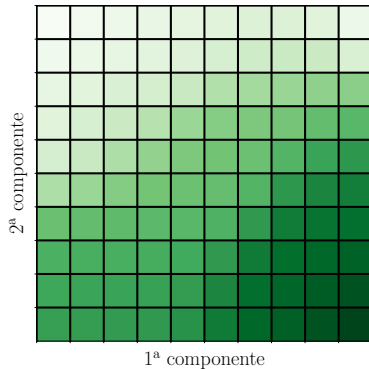
$$\mathcal{V}(\mathbf{a}, \mathbf{b}; \sigma) = e^{-\frac{\|\mathbf{a} - \mathbf{b}\|^2}{2\sigma^2}}$$



Clustering via SOM (partitioning based on CL method)

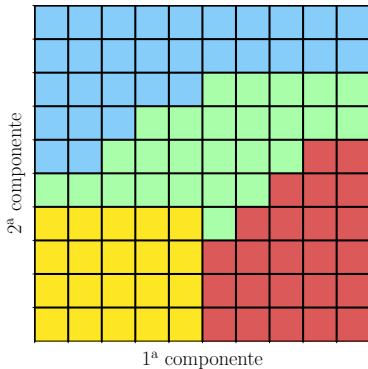


(o) Map of weights for first attribute

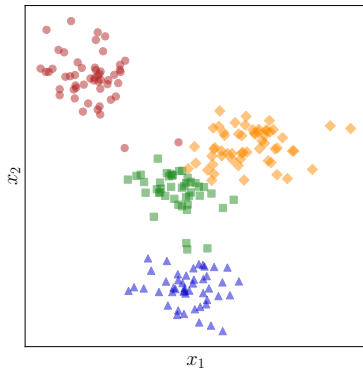


(p) Map of weights for second attribute

Clustering via SOM



(q) Neuron map response (using CL)



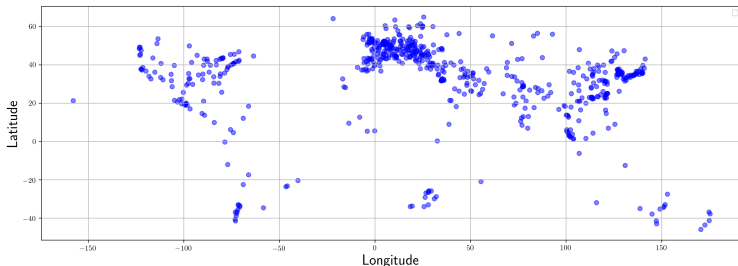
(r) Post-interpreted clustering

Agenda

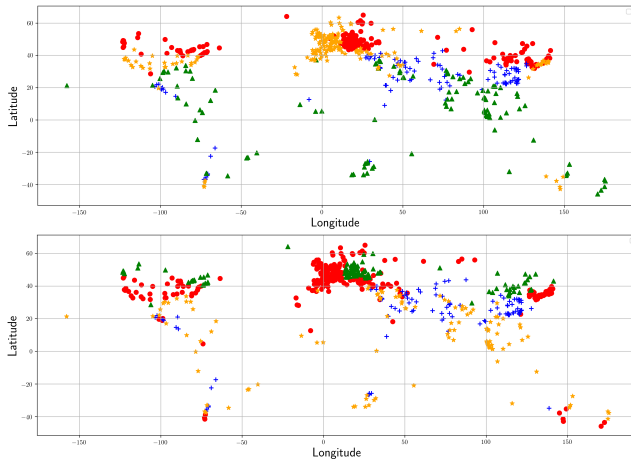
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Dataset (World Air Quality Index)

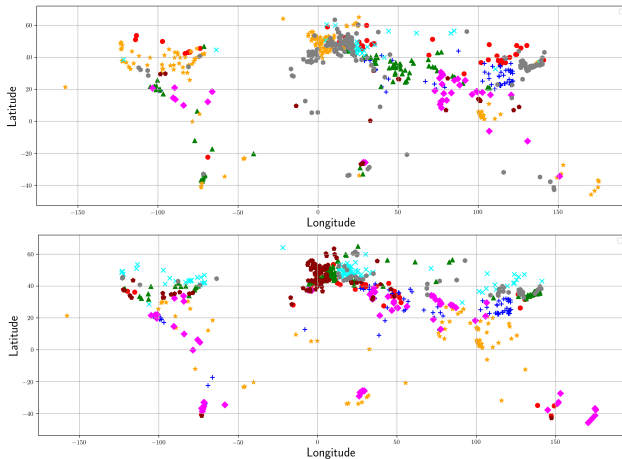
- Air quality dataset – Air Quality Open Data Platform – Worldwide COVID-19 dataset (data around March 15, 2020)
- Variables:
 - Latitude – Longitude – wind-speed (m/s) – wind-gust (m/s)
 - humidity (%) – dew point ($^{\circ}C$) – pressure (mb - millibars)
 - precipitation (inch 1.0 *inch* ≈ 25.4 *mm*) – temperature ($^{\circ}C$)
 - CO - carbon monoxide – SO₂ - sulfur dioxide – NO₂ - nitrogen dioxide – O₃ - ozone ($\mu g/m^3 \rightarrow$ EPA Standard)
 - pm10 - respirable particles – pm2.5 - fine particulate matter ($\mu g/m^3 \rightarrow$ EPA Standard)



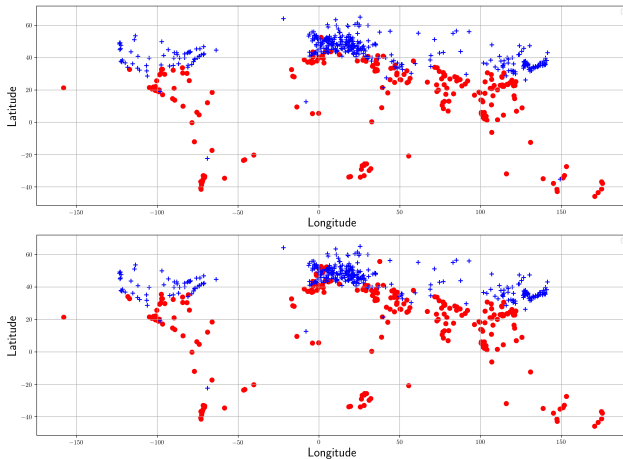
$c = 4$ – K -Means (top) – Ward (bottom)



$c = 8$ – GMM (top) – Fuzzy K -Means (bottom)



$c = 2$ – K -Means (top) – Fuzzy K -Means ($\beta = 2$) (bottom)



Clustering Methods (Hierarchical, Sequential, Mixture Models, K -Means, Fuzzy K -Means, and Kohonen Self-Organizing Maps)

rogerio.negri@unesp.br